

SLR ESTIMATES SUMMARY

Interested Quantity	Estimate	Std. Error	Test $H_0 : \beta = a$	Interval Estimate	Code
Noise variance σ^2	$\hat{\sigma}^2 = \frac{SS_{Res}}{n-2}$	—	—	$\left(\frac{SS_{Res}}{\chi_{n-2, \alpha/2}^2}, \frac{SS_{Res}}{\chi_{n-2, 1-\alpha/2}^2} \right)$	—
Slope β_1	$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$	$se(\hat{\beta}_1) = \sqrt{\hat{\sigma}^2 / S_{xx}}$	$\frac{\hat{\beta}_1 - a}{se(\hat{\beta}_1)} = t \stackrel{H_0}{\sim} t_{n-2}$	$\hat{\beta}_1 \pm t_{n-2}(\alpha/2) se(\hat{\beta}_1)$	<code>confint(reg, parm="x")</code>
Intercept β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$	$se(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}$	$\frac{\hat{\beta}_0 - a}{se(\hat{\beta}_0)} = t \stackrel{H_0}{\sim} t_{n-2}$	$\hat{\beta}_0 \pm t_{n-2}(\alpha/2) se(\hat{\beta}_0)$	<code>confint(reg, parm=1)</code>
<i>Parameter:</i> mean response at x_0 $\mu_{y x_0} = \beta_0 + \beta_1 x_0$	$\hat{\mu}_{y x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$	$se(CI) = \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$	—	Confidence interval: $\hat{\mu}_{y x_0} \pm t_{n-2}(\alpha/2) se(CI)$	<code>predict(reg, newdata, interval="confidence")</code>
<i>Random variable:</i> new individual at x_0 $y(x_0) = \beta_0 + \beta_1 x_0 + \epsilon$	$\hat{y}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$	$se(PI) = \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]}$	—	Prediction interval: $\hat{y}(x_0) \pm t_{n-2}(\alpha/2) se(PI)$	<code>predict(reg, newdata, interval="prediction")</code>

- We assume the simple linear regression model $y = \beta_0 + \beta_1 x + \epsilon$ with $\epsilon \sim N(0, \sigma^2)$.
- The terms SS_{Res} and S_{xx}/S_{xy} are defined by

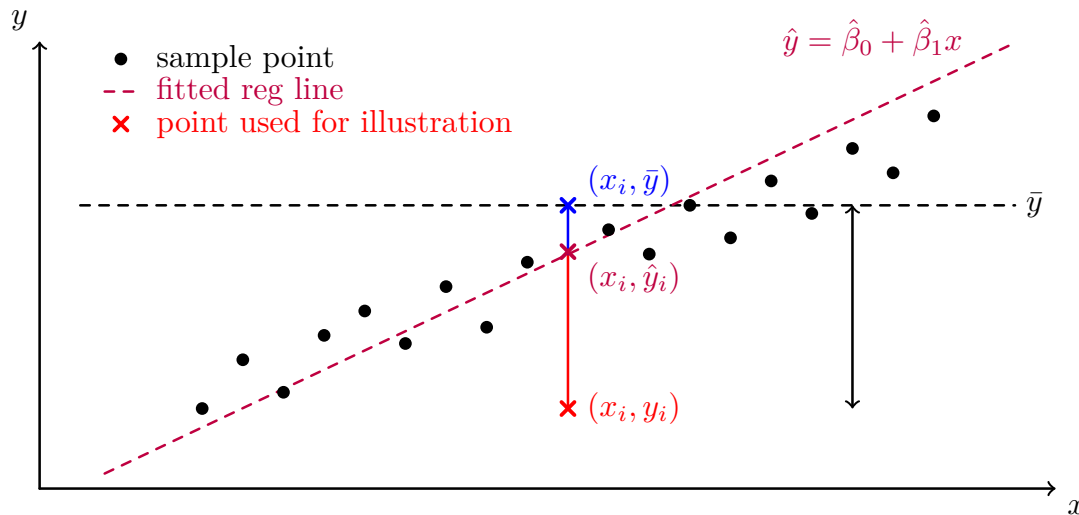
$$SS_{Res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \text{ and } S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

- In the Code column, `reg` is your model object, say `reg=lm(y~x)`, and the `x` in `parm=x` is your regressor name.

SLR ANOVA TABLE

Source	df	Sum of Square (SS)	Mean Square (MS)	F Statistic ($H_0 : \beta_1 = 0$)
Regression (x)	1	$SS_R = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$	$MS_R = SS_R/1$	$F = MS_R/MS_{Res} \stackrel{H_0}{\sim} F_{1,n-2}$
Residual (Res)	$n - 2$	$SS_{Res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$	$MS_{Res} = SS_{Res}/(n - 2)$	
Total	$n - 1$	$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2$		

- Remember the identity $SS_T = SS_{Res} + SS_R$. The following figure serves as an illustration (but not a rigor math proof).



Our goal is to explain the variation in y , measured by $(y_i - \bar{y})$, represented in the figure by $\uparrow$.

- Our fitted regression model gives us a prediction $\hat{y}_i$.
- This explains $(\hat{y}_i - \bar{y})$, represented in the figure by $\uparrow$.
- The $(y_i - \hat{y}_i)$ is unexplained, denoted in the figure by $\uparrow$.

So total variation $\uparrow =$ explained part $\uparrow$ + unexplained part $\uparrow$. Equivalently, SS_T (Total) = SS_R (Regression) + SS_{Res} (Residual).